

SAMYAK JAIN

Senior .NET Core Full Stack Engineer

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PROFESSIONAL SUMMARY

Senior Software Engineer with 10+ years of experience designing and delivering enterprise-grade backend systems, REST APIs, and microservices architectures. Strong background in Python-based services that mirrors .NET Core patterns including modular microservices, FastAPI ASP.NET Core REST design, Docker/Kubernetes deployments, AWS cloud infrastructure, and CI/CD pipelines. Hands-on with React.js frontends, SQL databases (MySQL/PostgreSQL/MS SQL), and production monitoring via Prometheus & Grafana. Skilled at modernizing legacy systems, performance tuning, and mentoring cross-functional teams in Agile (Scrum/Kanban) environments.

TECHNICAL SKILLS

Backend	C# / .NET Core / ASP.NET Core, Python, FastAPI, REST API Development, Microservices Architecture, gRPC (WCF-to-gRPC migration), OpenAPI / Swagger
Frontend	React.js, HTML5, CSS3, JavaScript
Databases	MS SQL Server, MySQL, PostgreSQL, MongoDB, DynamoDB, Redis / ElasticCache
Cloud & DevOps	AWS (EC2, S3, RDS, SNS, SQS, CloudWatch, IAM), Docker, Kubernetes, Terraform, Jenkins, Ansible, CI/CD
Monitoring	Prometheus, Grafana, CloudWatch, PerfView, dotnet-trace, dotnet-dump, LangSmith
Testing	xUnit / NUnit, Integration Testing, Testcontainers, Unit Testing Best Practices
OS & Tools	Linux Administration, Windows Services, IIS, PowerShell, Shell Scripting, Active Directory, Git / GitHub
AI / ML	LLM Integration (OpenAI, Claude), RAG Pipelines, Langchain, LangGraph, NLP, TensorFlow, PyTorch
Methodologies	Agile (Scrum / Kanban), SDLC, System Design, Technical Documentation, Team Leadership

WORK EXPERIENCE

TECH INFOSKY PVT. LTD.

Senior Software Engineer

Nov 2024 – Present

- Architected and developed modular microservices using FastAPI (Python) patterns directly transferable to ASP.NET Core microservices enabling independent scaling across enterprise systems.
- Designed and deployed RESTful APIs (OpenAPI/Swagger-documented) consumed by web dashboards and mobile clients, with secure authentication and role-based access control.
- Built NLP-powered data extraction pipelines to parse unstructured text from emails and reports similar to enterprise document processing in .NET environments.
- Implemented Redis (ElasticCache) caching layer to reduce API response latency during peak load, analogous to IDistributedCache patterns in ASP.NET Core.
- Set up Prometheus + Grafana monitoring dashboards for real-time performance visibility and proactive incident response equivalent to Application Insights / PerfView workflows.
- Led code reviews, mentored junior developers, and enforced SDLC best practices (branching strategy, PR standards, integration testing) across the team.
- Managed Docker-containerized deployments on AWS (EC2, S3, RDS) with CI/CD pipelines for automated build-to-production workflows.

ENTERPRISE MINDS (FLIPKART)

Software Engineer

Jul 2024 – Oct 2024

- Developed and maintained REST APIs enabling real-time data exchange between microservices consistent with ASP.NET Core Web API design patterns.
- Optimized MySQL/PostgreSQL schemas (indexing, query tuning) to minimize latency during high-traffic periods skills directly applicable to MS SQL Server performance work.
- Managed AWS cloud infrastructure (EC2, RDS, S3) with auto-scaling policies for high availability and cost optimization.
- Automated server provisioning and maintenance using Shell scripts on Linux/Unix, reducing manual deployment effort significantly.
- Implemented modular backend service design with clear separation of concerns, mirroring Clean Architecture principles used in enterprise .NET solutions.

CIPLA

Software Engineer

Aug 2022 – Jun 2024

- Built REST APIs and event-driven backend services integrated with CloudWatch and SNS for monitoring and alerting experience aligned with Azure Service Bus / RabbitMQ patterns.
- Developed predictive analytics models using TensorFlow/PyTorch for customer behaviour forecasting, deployed as microservices with Docker and Kubernetes.
- Designed AWS QuickSight dashboards for business stakeholders, providing data-driven insights without manual database querying.
- Used Ansible and Terraform for infrastructure-as-code provisioning and multi-server configuration experience applicable to Bicep / ARM template workflows.
- Worked within Scrum/Kanban frameworks with sprint planning, daily standups, and cross-functional collaboration to deliver production features on schedule.

NATCO PVT. LTD.

Associate Software Engineer

Sep 2015 – Jul 2022

- Developed RESTful APIs using Python/FastAPI for web and mobile apps foundational REST design experience equally applicable to ASP.NET Core Web API.
- Managed structured data using MySQL and PostgreSQL with proper indexing, query optimization, and schema design directly applicable to MS SQL Server environments.
- Set up role-based IAM access controls on AWS to restrict sensitive cloud resource access mirrors Active Directory / Windows Service identity management.
- Deployed backend systems using Docker and Jenkins CI/CD, automating code-to-production pipelines with minimal manual intervention.
- Participated in Agile ceremonies (sprint planning, standups, client demos) to deliver features with quality and on time.

KEY PROJECTS

AI-Powered Product Intelligence & Support System (E-commerce) | Nov 2024 – Present

- Designed a multi-agent LLM system using LangGraph — architecture pattern analogous to orchestrated .NET microservice pipelines reducing customer support load by over 60%.
- Built FastAPI microservices (REST + async) deployed on AWS (EC2, S3, RDS, DynamoDB), monitored with CloudWatch and Grafana; same deployment model as ASP.NET Core on AWS.
- Integrated RAG pipelines with vector databases (FAISS, ChromaDB) for context-aware responses; reduced AI hallucinations by 35% through LangSmith-based prompt testing.

Tech: FastAPI, Python, AWS, Docker, CI/CD, Grafana, LangSmith, FAISS, ChromaDB

Dynamic Pricing & Demand Forecasting System (Hotel Chain) | Aug 2022 – Jun 2024

- Built real-time pricing microservices using Python (Django + Flask), deployed with Docker and Kubernetes; Terraform used for IaC patterns identical to .NET Core Kubernetes deployments.
- Trained PyTorch ML models for demand forecasting with 87% accuracy; implemented predictive analytics that increased off-season occupancy by 22%.
- Integrated AWS services (S3, RDS, SNS, SQS) for event-driven communication experience applicable to Azure Service Bus / RabbitMQ message queue architectures.

Tech: Python (Django, Flask), PyTorch, MongoDB, PostgreSQL (RDS), AWS, Docker, Kubernetes, Terraform, CloudWatch

Guest Experience & Feedback Automation System | Sep 2015 – Jul 2022

- Designed GenAI-powered chatbots and NLP-based sentiment analysis pipelines for hotel/restaurant chains, deployed via Docker and Kubernetes with continuous delivery.
- Implemented AWS SNS/SQS alerting and routing for critical incident escalation — directly applicable to enterprise event-driven architectures and on-call incident management.

Tech: Python (Flask), GenAI, PyTorch, NLP, MongoDB, DynamoDB, AWS, Docker, Kubernetes, Terraform, Grafana

EDUCATION

Bachelor of Technology (B.Tech) Oriental College, Bhopal, M.P.	2015
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LANGUAGES

English (Professional) **Hindi** (Native) **Marathi** (Working Proficiency)